Vessel Target Detection Method Based on an Improved CFAR Method in Nighttime Remote Sensing Images

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***Abstract*— The Visible Infrared Imaging Radiometer Suite (VIIRS) day/night band (DNB) data are very sensitive to low radiation and capable of detecting faint light sources emitted by vessels at night. Existing vessel detection methods are mainly based on pixel statistical feature estimation or manual experience to determine the threshold for detection. In complex scenes, problems such as interference targets and inaccurate background modeling result in inaccurate detection accuracy and poor robustness. In this article, a vessel detection method based on an improved constant false alarm rate (CFAR) detector is proposed, and more accurate threshold selection is obtained through background sample truncation and distribution adjustment. Then, the local peak detection algorithm removes the bright spots in the nonvessel location and realizes the vessel detection. We selected data from two research areas, the waters around the East China Sea and the waters around the Gulf of Mexico of the United States, to build a vessel dataset and applied the algorithm to the dataset. The experimental results show that the total detection accuracy of the algorithm is 94.48%, and the recall rate is 93.40%, which is higher than the other two comparison methods. Especially in complex scenarios with high target density, such as ports, the recall rate is significantly improved, which proves the applicability of the algorithm.**

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I. Introduction

MARINE monitoring is of great significance in safeguarding maritime rights and strengthening the supervision of naval areas [1], [2]. Remote sensing technology can simultaneously observe the sea surface in large areas [3]. Thus, its role in marine monitoring is becoming more and more critical. However, traditional optical remote sensing relies on the detection of reflected light from the Sun and cannot provide effective nighttime data. As a new direction and cutting-edge hotspot in the development of remote sensing, low-light remote sensing is capable of detecting weak light sources on the surface at night or dusk, compensating for the limitations of traditional optical remote sensing and providing nighttime radiation products [4], [5]. Several relevant studies have already proved the potential of low-light remote sensing images to detect luminous vessels at night [6], [7]. Therefore, low-light remote sensing technology has become a brand-new means of nighttime vessel detection and can be an effective supplement to traditional remote sensing technologies, such as SAR and infrared remote sensing in the task of nighttime marine monitoring [8], [9], [10].

However, as shown in Fig. 1, the following issues need to be overcome to successfully utilize low-light remote sensing data for nighttime marine vessel detection.

1) Low-light remote sensing imagers mainly detect artificial light sources on the surface to form radiometric products [11], while the spatial resolution of the low-light remote sensing data is relatively low, which leads to the display of the vessel target in low-light remote sensing data as a "bright spots," i.e., pixels or small areas with higher brightness compared to the surrounding background, and, therefore, cannot provide effective contours, textures, and other relevant features that can be used for vessel detection and identification.
2) The brightness of vessel lights in remote sensing images is affected by observation angle, atmospheric conditions, sensor noise, and other factors on the one hand [12] and by random factors of the vessel itself such as vessel functions, light settings, and operational status on

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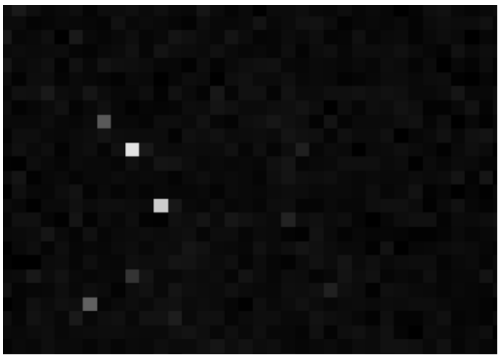


Fig. 1. Sample of the vessel target in VIIRS DNB data.

the other hand. As a result, the brightness of different ships can vary greatly, and there is almost no reliable prior knowledge to determine the specific brightness of vessels' light sources.

3) Low-light imagers have characteristics including high gain, poor signal-to-hybrid ratio, and significant influence by stray light, which lead to the complex background and noise environment in the remote sensing image to cause serious interference to the vessel target and create more significant difficulties for the vessel target detection task.

Considering that the brightness of marine vessel lights at night is significantly higher than other light sources in the localized area, many researchers have designed their detection method based on threshold detectors. In response to the previously mentioned characteristics of low-light remote sensing data that negatively affect the effectiveness of vessel detection, researchers mainly proposed two kinds of solutions: improving data quality through preprocessing and designing adaptive detection thresholds based on local statistical data. In addition, a few researchers have attempted to apply target detection methods based on deep learning techniques to this field. These works will be introduced separately next.

Many researchers suppressed the background brightness and enhanced the target brightness by controlling the noise and enhancing the contrast in the preprocessing stage or established empirical models to compensate for the effects of natural phenomena on the target brightness in order to improve the data quality and highlight the vessel's lighting information. For example, Elvidge et al. [13] used a regional adaptive Wiener filter in the preprocessing stage to reduce the stripe noise of remote sensing images and used logarithmic computation to enhance the contrast and highlight locally brighter vessel targets. They designed the spike median index (SMI) in the vessel identification stage, which further suppressed the light diffusion and background noise by subtracting the regional median value to obtain a high-quality image for detection. Elvidge et al. [8] further improved their work by using gradient transform in the preprocessing stage to obtain images that characterize local variations and obtain clearer and high-quality images. Based on the SMI threshold method, Sun et al. [14] added preprocessing steps such as geometric correction and radiation enhancement to further improve the detection accuracy. Kim et al. [15] consider the effect of the moon phase on the brightness of vessels and calculate the offset for brightness correction by designing an empirical model in the preprocessing stage to eliminate the impact of the moon phase on the image quality. Xue et al. [16] discussed the effects of attenuation and diffusion on image quality due to atmospheric transmission of vessels' lights. They quantitatively analyzed the diffusion by building an empirical model to compensate for the loss of the vessels' light source.

Some researchers have analyzed and modeled the local background noise and other relevant situations in the detection region and designed variable thresholds for detection to cope with complex scenes. Duan et al. [17] used the average brightness value of the points corresponding to several maximum gradients in the gradient map to determine the threshold value. Ganggang et al. [18] designed a threshold selection strategy called the maximum entropy method (MaxEnt). The threshold is calculated to maximize the entropy sum of the target and the background according to probabilities of pixel radiances indicated by a histogram. Then, a local spike detection (LSD) algorithm is proposed to remove the false identification pixels near the target vessel pixels illuminated by the vessel lights. Lebona et al. [19] and Liang and Xiaosheng [20] applied the constant false alarm rate (CFAR) [21] detector to DNB data. The algorithm uses the sample data in the background window for parameter estimation to characterize the local background and noise conditions, and the threshold is set by keeping the proportion of the number of background samples above the threshold constant, which is the false alarm rate. Pixels above this threshold are considered unusually bright and, therefore, likely to be samples from the target.

Some researchers applied deep learning methods to extract relevant features and enhance the robustness of detectors automatically. For example, Shao et al. [22] proposed an improved YOLOv5 [23] algorithm and named it TASFF YOLOv5. The algorithm is designed based on multiscale feature fusion to enhance the detection performance of small targets. Using the hierarchical characteristic of the convolutional neural network, the authors designed a four-layer spatial feature fusion network for vessel detection. Motomura and Nagao [24] designed FishNet based on the image segmentation network U-net [25] and added temporal feature inputs to roughly estimate the number of vessels in the area. Wantao et al. [26] added the image scale expansion and attention mechanism module to the Fast R-CNN network [27] to screen the characteristics of luminous vessel targets from the channel and spatial dimensions. They proposed this improved Fast R-CNN method called scale expansion attention fusion fast (SEAFF) R-CNN for vessel detection. Zuo et al. [28] established an enhanced dense nested-attention network (DNA-net) to improve the detection of small vessel targets under low-light conditions. This approach effectively integrates the original VIIRS/DNB, SMI, and spike height index (SHI) images to maintain deep-level features and enhance feature extraction.

The current research on low-light vessel detection has achieved positive results, but there are still many problems that need to be improved.

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1) Feature enhancement methods can improve the quality of low-light remote sensing data under ideal conditions, but they have the opposite effect in complex situations. For example, researchers use logarithmic computation or SMI to control noise and enhance image contrast. However, considering the large differences in the radiance of different vessels and the effects of complex atmosphere and noise conditions on vessels' lights in different regions, it is impossible to ensure that these methods can enhance rather than suppress all vessels' lights in complex scenes. The empirical models established by researchers based on the principles of natural phenomena usually need atmospheric inversion products or other reference data; these reference data may be missing or inaccurate. Also, these empirical models are typically designed for ideal conditions, which makes it challenging to overcome the influence of complex noise and abnormal background.
2) Variable threshold-based methods suffer from the problem of being affected by interfering targets and inaccurate background modeling and perform poorly in scenes with complex backgrounds and dense targets such as busy harbors or shipping lanes. The selection of variable thresholds usually relies on statistical modeling of the background noise situation in the region around the target point. In scenarios such as busy harbors or shipping lanes, anomalies and interfering targets with much higher radiance than other background points may appear in the area around the target point, which may lead to contamination of the background statistical samples, resulting in inaccurate parameter estimation and modeling, and ultimately leading to serious degradation of the performance of the vessel detector. In addition, the empirical models or Gaussian distributions usually used by current researchers cannot effectively characterize the distribution of background and random noise in the relevant region, which affects the accurate selection of thresholds.
3) Current deep learning methods do not apply to the task of recognizing vessels' lights that lack relevant features. Low-light remote sensing imager detects vessel lights in order to form a radial product, which lacks relevant features such as contour and texture. The deep learning idea is currently not effectively adapted based on the actual situation, failing to obtain convincing results while consuming a large amount of computational resources.

Based on the consideration of the current research status, in this study, we developed a vessel detection method for the Visible Infrared Imaging Radiometer Suite (VIIRS) [29] Day/Night Band (DNB) [30] data of the Suomi National Polar-orbiting Partner (SNPP) and Joint Polar Satellite System (JPSS) satellite [31]. We created a dataset of 862 vessel targets with vessel data from different sea areas to ensure the diversity of images in the dataset. We design a complete unsupervised vessel detection algorithm. This detection method is based on the method proposed by Liang and Xiaosheng [20] with tremendous adjustments and improvements for vessel detection to cope with the actual situation of DNB data. Considering that anomalies and interfering targets in the background window may seriously interfere with the estimation of the background distribution parameters and affect the threshold value, the median variability index (MVI) is designed and utilized to achieve the truncation depth of the samples iteratively and ultimately obtain the undisturbed background samples and the adjusted Weibull distribution. Subsequently, the adjusted Weibull distribution is used to model the background samples, and the threshold is calculated using the false alarm rate to realize the target detection. Finally, the nonvessel position bright spot is removed by LSD, and the vessel position information is accurately detected from the image. Experiments show that this method performs well in our dataset, especially in complex scenes with high target density, such as busy harbors, and outperforms current mainstream detection methods.

The rest of the article is organized as follows. Section II presents the technical details related to DNB data from SNPP and JPSS-1 satellites. Section III provides a detailed description of the vessel detection method designed in this article. Section IV presents the relevant details of the dataset used for the experiments, as well as the experimental results of our detection method. Section VI draws conclusion.

II. Data

A. VIIRS DNB Data

The VIIRS is carried by the SNPP and JPSS-1 satellites. The SNPP and JPSS-1 satellites operate in Sun-synchronous orbits of about 824 km. As the crucial instrument of these two satellites, VIIRS is a scanning radiometer with a side-scanning width of about 3044 km and a revisit period of 12 h, with a wavelength range of 0.41–12.50 μm. VIIRS has five high-resolution imaging bands (I-band), 16 medium-resolution bands (M-band), and a panchromatic band (DNB), a total of 22 bands [32].

DNB is a panchromatic band covering 500–900 nm in spectral space with a spatial resolution of 750 m and a wide dynamic range of about seven orders of magnitude [33]. This allows the DNB imager to observe the Earth's surface from day to night. The DNB imager uses three gain stages to achieve the wide dynamic range, namely, the low gain stage (LGS), the medium gain stage (MGS), and the high gain stage (HGS). The three gains are used to observe the daytime land surface, the area near the terminator line, and the nighttime land surface, respectively. In order to increase the signal, the time delay integration (TDI) of the three gain stages has 1, 3, and 250 pixels, respectively [34]. Technical details about the VIIRS instrument and DNB data are shown in Table I. This study used the VIIRS/DNB sensor data record (SDR) dataset, which contains two types of data: 1) the scientific data record (SVDNB) providing the radiance of NTL and 2) the geolocation content (GDNBO) that supplies the corresponding longitude, latitude, satellite viewing zenith angle, lunar zenith angle (LZA), and lunar phase angle (LPA). All data were from the NOAA's comprehensive large array data steward system.

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TABLE I
TECHNICAL DETAILS ABOUT THE VIIRS INSTRUMENT AND DNB DATA

	Index Item	Detail
VIIRS	Orbit height	~842km
	Imaging width	~3044km
	Revisit cycle	12h
	Bandwidth	0.41~12.50μm
	Band	5 I Band, 16 M Band，1 DNB 14 RSBs，7 TEBs
DNB	Pixel resolution	~742m
	Detection spectrum	500~900nm
	Dynamic range	$0.02\sim3\times10^{-9}\mathrm{W}\cdot\mathrm{cm}^{-2}\cdot\mathrm{sr}^{-1}$
	SNR	>10 (Prelaunch Performance) >9 (On-Orbit Performance)

B. Vessel Automatic Identification System Data

Automatic identification system (AIS) is composed of shore-based (base station) facilities and vessel-borne equipment, which is a new type of digital navigation aid system and equipment integrating network technology, modern communication technology, computer technology, and electronic information display technology. The proper use of AIS helps to enhance the safety of life at sea, improve the safety and efficiency of navigation, and protect the Marine environment.

The AIS data transmitted by vessels are received and recorded by the base station, and these AIS data record some status information of the vessel itself at that moment. Although AIS data cannot cover all vessels at sea for vessel detection tasks, it can provide accurate time, location, and category information for some vessels in a certain area. Since the vessels are subpixel targets in DNB image data, which cannot be determined by manual annotation alone, therefore, this article uses AIS data to assist in building a low-light remote sensing vessel dataset for detection and verification.

III. METHOD

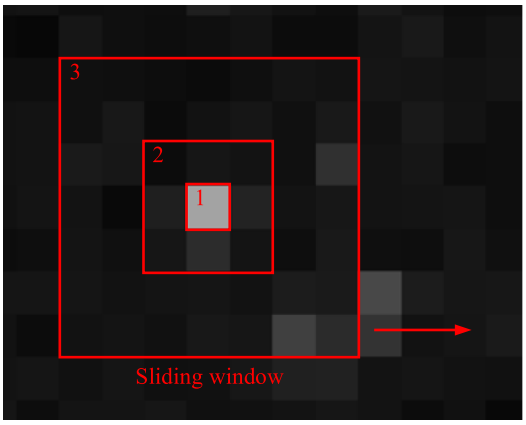
A. CFAR Method for Vessel Detection

The spatial resolution of the DNB remote sensing image is only 750 m, so vessel targets are presented in the image as bright spots, usually occupying only one pixel. We summarize the vessel target detection problem as a hypothesis testing problem. For a certain target point p to be detected, there are two possible hypotheses, H_1 and H_0, respectively,

$$H_1 = t(x) + n(x) \tag{1}$$

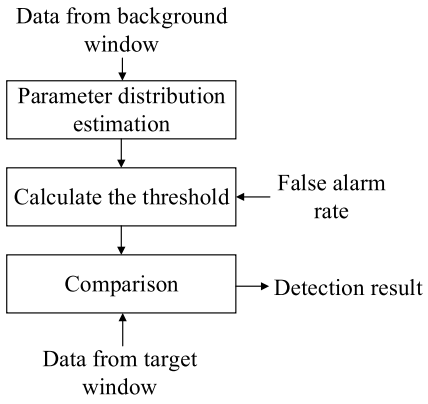
$$H_0 = n(x) \tag{2}$$

where H_1 and H_0 represent the hypothesis of the presence and absence of the target, respectively, where $n(x)$ represents the background component and $t(x)$ represents the target component. It is obvious that these two random variables have different distributions and are independent of each other. If the prior knowledge of both the background and the target is known, the optimal detector can be designed under the guide of the Neyman–Pearson criterion [35].



1. target window 2. background window 3. protection window

(a)



(b)

Fig. 2. Structure diagram of the 2-D CFAR detector. (a) Sliding window. (b) Principle flow of CFAR detection.

However, as we mentioned in Section I, the radiance value of the vessels in the low-light remote sensing image is affected by many uncontrollable random factors, such as observation angle, atmospheric conditions, and the vessel's own light settings, sailing status, and so on. Even the same vessel target may have different radiance values under different conditions. In addition, vessels with different types and functions have different light source settings and obviously different radiance values. Therefore, the prior knowledge of vessel radiance is difficult to obtain. We consider the use of a suboptimal scheme, the CFAR detector, which automatically adjusts the determination of the detection thresholds based on accurate modeling of the statistical distribution of the background clutter power or the noise situation, ensuring that the false alarm rate is close to the set expectation. They are usually implemented using the sliding window technique, and the reference samples in the local reference window are used to estimate the parameters of the distribution model.

Fig. 2 shows the structure diagram of the 2-D CFAR detector [21]. The detection task is implemented based on a sliding processing window, which consists of a target window, a background window, and a protection window. Ideally, the target itself is contained in the protection window, and the protection window acts as an isolation to eliminate the impact of the distribution parameter estimation of background noise caused by the leakage of the target light source into the

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surrounding pixels. The pixels in the background window are considered background samples and are used to estimate parameters. Let the set of N sample pixels in the background window be $X_b = [x_1, x_2, \ldots, x_N]$, and assume that the background pixel prior distribution is known. The associated parameter estimates of the background distribution obtained based on the set of samples X_b can be denoted as $\hat{\theta}(X_b)$. Thus, the decision rule of the CFAR detector is

$$T\left[X|\hat{\theta}(X_b)\right] = \frac{x_0}{\hat{\theta}(X_b)} \underset{H_0}{\overset{H_1}{\gtrless}} \tau \tag{3}$$

where $X = [x_0, x_1, x_2, \ldots, x_N]$, x_0 is the interest target sample to be decided, and H_1 and H_0 represent the hypothesis that the target exists and does not exist, respectively. $T[X|\hat{\theta}(X_b)]$ represents the value of the target sample to be tested. τ is the threshold calculated under the background distribution condition $\hat{\theta}(X_b)$ and the preset false alarm rate P_{FA}. τ is given by

$$\begin{aligned} P_{\mathrm{FA}} &= E\left\{P\left(T\left[X|\hat{\theta}(X_b)\right] > \tau|H_0\right)\right\} \\ &= \int_0^{+\infty}\int_{\tau}^{+\infty} f_T\left(t|\hat{\theta}, H_0\right) f(\hat{\theta}) dt d\hat{\theta} \end{aligned} \tag{4}$$

where $E[\cdot]$ denotes expectation and $P[\cdot]$ denotes probability. $f_T(t|\hat{\boldsymbol{\theta}}, H_0)$ is the probability density function (pdf) of $T[X|\hat{\theta}(X_b)]$ under condition H_0, and $f(\hat{\theta})$ is the pdf of $\hat{\theta}(X_b)$. It is clear that the key to designing a CFAR detector is to analyze the distribution of background composition, find a suitable method to estimate the parameters of the background distribution, and then calculate the threshold τ based on the false alarm rate P_{FA}.

B. Improvement Ideas for Detection Algorithm

In practical applications, in areas such as busy shipping lines and crowded ports, anomalies and interfering targets in the background window may seriously interfere with the estimation of the background distribution parameters and affect the threshold value. These anomalies and interfering targets usually have higher radiance values than other data points in the background region. Generally, when there are one or more interfering points in the background window in the form of nonocean targets or their sidelobes or ghosting, both the mean and variance of the samples in the background window will increase, leading to an elevated detection threshold. This makes the observed false alarm probability drop below the preset value and reduces the probability of detection, which is called the capture effect [36], [37]. From previous studies, there are two main solutions: 1) removing the interfering samples from the background samples and 2) characterizing the background with modeling methods that are less affected by interfering samples, such as methods like OS-CFAR [38]. It is obvious that method (2) cannot guarantee the complete elimination of the influence of interfering samples, while only a small portion of the sample set is used for parameter estimation and decision-making, which affects the accuracy of detection.

Therefore, we designed a CFAR detection method based on data truncation for vessel detection with low-light remote sensing data, which can overcome the effects of interfering samples and obtain satisfactory detection accuracy in complex areas. We used the median variation index (MVI) to eliminate interference items for the sorted sample sequence iteratively. Then, we conducted rigorous statistical analysis on the truncated data and used the adjusted Weibull distribution for parameter estimation and background modeling. Next, we completed threshold selection and detection with a false alarm rate. Finally, the nonvessel position bright spot is removed by LSD, and the vessel targets are accurately detected from the image. The specific idea is shown in Fig. 3. The detailed execution principle of threshold calculation and comparison steps is shown in (3) and (4). The details about interference data removal, distribution adjustment, parameter distribution estimation, and nonvessel bright pixel removal will be presented in Sections III-C–III-E.

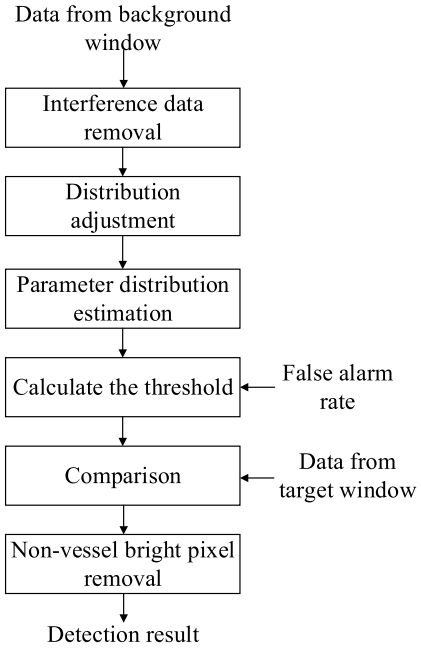


Fig. 3. Process diagram of our vessel detection method.

C. Interference Data Removal

The idea of eliminating the influence of interference background samples on parameter estimation is to eliminate the background samples in the background window. Considering that these interference samples are significantly brighter, some researchers designed to remove the maximum value of a fixed number in the sample set after sorting [39] or set a threshold and remove samples greater than the threshold [40]. The relevant parameters of these methods are preset and kept constant in the experiments, and the setting of the parameters relies on manual experience and a priori information, which leaves room for further improvement in complex scenarios.

Therefore, we design algorithms to accurately and automatically identify interference samples in the sample set of background distribution parameter estimates. The algorithm details are given in Algorithm 1. First, the reference samples in the background window are sorted to form the sorted

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sample set $E_x = \{X(1) \leq X(2) \leq \cdots \leq X(k)\}$. Then, the ranked sample set is reviewed in order, and the outliers are removed from the sorted sample set one by one through an iterative hypothesis testing method based on the variability index value of the sample set. Therefore, we need an index that can represent the existence of outliers in the sample set. We expect that the index value can show a large difference between the stationary dataset without outliers and the dataset containing them. At the same time, the value of the index can show a significant and regular change with the gradual elimination of outliers. We designed the MVI for the above related requirements, which can be expressed as

$$\mathrm{MVI} = \frac{|\mu_x - x_{\mathrm{median}}|}{\mathrm{MAD}} \tag{5}$$

where μ_x is the mean value of the sample set $E_x = \{X(1) \leq X(2) \leq \cdots \leq X(k)\}$ and x_{median} represents the median value of the sample set. MAD is the median absolute deviation, which can be expressed as

$$\mathrm{MAD} = \mathrm{median}(|x_i - x_{\mathrm{median}}|) \tag{6}$$

where x_i is a sample of the sample set E_x. $i = 1, 2, \ldots, k$ median stands for the operation that takes the median.

In fact, the distribution of interference targets or outliers is sparse. In the process of iteratively eliminating the largest sample, a slight reduction in the number of sample sets has little impact on the value of MAD, making the MVI value relatively stable in the iterative process. However, considering that the radiance value of the interference target is much larger than that of the background, the mean value will be significantly smaller in the iterative process, which leads to a change in the derivative of the MVI curve. We designed the interference item elimination algorithm based on the above ideas, and the specific details are shown in Algorithm 1.

Algorithm 1 Interference Background Sample Removal

Input: Sliding window data E_x, Threshold λ
Initialize: Sort elements in E_x, $E_x = \{X(1) \leq X(2) \leq \cdots \leq X(N)\}$, $k = N$
While E_x **not empty:**
  Sort elements in E_x, $E_x = \{X(1) \leq X(2) \leq \cdots \leq X(k)\}$
  Calculate the Median variability index $MVI(k)$ of E_x
  If $MVI(k) - MVI(k+1) > \lambda$
    End while
  End If
  Remove $X(k)$ from E_x
  Update k: $k = k - 1$;
End while
Output: Censored data E_x

D. Distribution Adjustment and Parameter Distribution Estimation

Now that we have eliminated the interference sample in the background sample set, we will use the reduced sample set for background distribution modeling. A common simplification is maintaining the original model assumptions after outliers are eliminated and applying standard estimators for the untruncated distribution to the reduced sample set, which obviously produces erroneous results. Specifically, the process of removing high-intensity outliers artificially interferes with the distribution of the sample set, which leads to the inevitable underestimation of the variance or scale parameter and, ultimately, a wrong threshold, unless this is considered in the model and, therefore, compensated in the estimator. We refer to the idea of Tao et al. [39], adjust the background distribution model based on the truncation result of the sample set, and use the adjusted distribution to model the background and noise. For the set of background random variables X, it is assumed that the pdf and the cumulative distribution function (cdf) of the following distribution are $p_X(x)$ and $P_X(x)$, respectively. The adjusted distribution can be defined as

$$p_{\tilde{X}}(x;t) = p_X(x|X<t) = \begin{cases} \dfrac{p_X(x)}{P_X(t)}, & 0 < x \leq t \\ 0, & x > t \end{cases} \tag{7}$$

where $\tilde{X}$ is the sample set after truncation; in this process, all variables with lower value than threshold t are eliminated from the sample set X. Corresponding to the outlier elimination algorithm stated in the previous section, for a sorted sample set containing k samples, if n outliers are eliminated, then $t = X(k-n)$.

The Weibull distribution provides a close approximation to the probability laws of many natural phenomena. In recent years, most attention has been focused on this method, not only due to its greater flexibility and simplicity but also because it can be a good fit for experimental data. Weibull distribution can be used to model right-skew, left-skew, or symmetric data. It has been widely used in the field of ocean monitoring and is capable of characterizing sea clutter, surface backscatter, and other situations. We use Weibull distribution to model sea surface background samples. The pdf of the Weibull distribution function is expressed mathematically as

$$f_X(x;a,b,c) = \begin{cases} \dfrac{c}{b}\left(\dfrac{x-a}{b}\right)^{c-1} \exp\left[-\left(\dfrac{x}{b}\right)^c\right], & x \geqslant a \\ 0, & x < a \end{cases} \tag{8}$$

where a, b, and c are position parameters, scale parameters, and shape parameters, respectively. According to the actual situation of low-light remote sensing data, we set the position parameter a in the Weibull distribution model to 0. The cdf can be obtained as

$$F_X(x;b,c) = 1 - \exp\left[-\left(\frac{x}{b}\right)^c\right], \quad x \geqslant 0. \tag{9}$$

Based on (7), the adjusted truncated distribution is

$$f_{\tilde{X}}(\tilde{x};b,c,t) = \begin{cases} \dfrac{c\tilde{x}^{c-1}\exp\left[-\left(\frac{\tilde{x}}{b}\right)^c\right]}{b^c\left\{1-\exp\left[-\left(\frac{t}{b}\right)^c\right]\right\}}, & 0 < \tilde{x} \leqslant t \\ 0, & \tilde{x} > t \end{cases} \tag{10}$$

where t is the truncation threshold and $\tilde{x}$ are samples for parameter estimation. For a sorted sample set, if l samples are left after sample set truncation, $X(l)$ is the maximum sample,

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$t = X(l)$. The log-likelihood function of the adjusted Weibull distribution can be obtained as

$$\ln L(b, c|\tilde{x}, t) = l(\ln c - c\ln b) + (c-1)\sum_{i=1}^{l}\ln \tilde{x}_i - \frac{1}{b^c}\sum_{i=1}^{l}\tilde{x}_i^c - l\ln\left[1 - e^{-\left(\frac{t}{b}\right)^c}\right]. \quad (11)$$

The partial derivative of b and c is obtained, and the maximum likelihood equation is

$$\begin{cases} \varphi_1 = \dfrac{\partial \ln L}{\partial b} = -\dfrac{lc}{b} + \dfrac{c}{b^{c+1}}\displaystyle\sum_{i=1}^{l}\tilde{x}_i^c - \dfrac{lct^c}{b^{c+1}\left[1 - e^{(t/b)^c}\right]} = 0 \\ \varphi_2 = \dfrac{\partial \ln L}{\partial c} = \dfrac{l}{c} - \ln b + \displaystyle\sum_{i=1}^{l}\ln \tilde{x}_i \\ \qquad - \displaystyle\sum_{i=1}^{l}\left(\frac{\tilde{x}_i}{b}\right)^c \ln\left(\frac{\tilde{x}_i}{b}\right) + \frac{\left(\frac{t}{b}\right)^c \ln\left(\frac{t}{b}\right)}{1 - e^{\left(\frac{t}{b}\right)^c}} = 0. \end{cases} \quad (12)$$

Given that this is a nonlinear system, we adopt the Newton iteration method to solve it. For a set of approximate solutions b_k and c_k of the equation set, after adopting the Taylor expansion method to the nonlinear functions φ_1 and φ_2 in the neighborhood of the point (b_k, c_k) and omitting the quadratic terms, the system can be linearized as follows:

$$A_k(\lambda - \lambda_k) + \Phi_k = 0. \quad (13)$$

Based on this linear equation, we iteratively solve the maximum likelihood estimators for b and c, where

$$A_k = \begin{bmatrix} \dfrac{\partial \varphi_1}{\partial b} & \dfrac{\partial \varphi_1}{\partial c} \\ \dfrac{\partial \varphi_2}{\partial b} & \dfrac{\partial \varphi_2}{\partial c} \end{bmatrix}_{\substack{b=b_k \\ c=c_k}}, \quad \lambda = \begin{bmatrix} b \\ c \end{bmatrix}, \ \lambda_k = \begin{bmatrix} b_k \\ c_k \end{bmatrix}$$

$$\Phi_k = \begin{bmatrix} \varphi_1 \\ \varphi_2 \end{bmatrix}_{\substack{b=b_k \\ c=c_k}}.$$

A_k is the Jacobian matrix of the maximum likelihood function L. If A_k is a nonsingular matrix, the iterative solution $\lambda_{k+1} = \lambda = \lambda_k - A_k^{-1}\varphi_k$. Thus, we can set initial values a_0 b_0 and iteratively solve the relevant parameters. The iteration continues until the convergence condition $|b_{k+1} - b_k| < \varepsilon_1$, $|c_{k+1} - c_k| < \varepsilon_2$ is satisfied. ε_1, ε_2 is the minimum error given in advance. The iterative solution algorithm is shown in Algorithm 2.

After obtaining the Weibull distribution parameter estimates for the background samples, we can perform the detection task by obtaining the threshold based on the principle of the CFAR detector

$$T\left[X|\hat{b}(x_i), \hat{c}(x_i)\right] \underset{H_0}{\overset{H_1}{\gtrless}} \tau. \quad (14)$$

E. Nonvessel Bright Pixel Removal

At present, we have distinguished the background pixel from the target pixel lit by the vessel through the detection algorithm. Still, in the actual situation, not all the target pixels lit by the vessel lights contain a vessel. For example, some vessels are equipped with high-power lighting equipment for fishing or other functions, and some vessels are located just near the boundary between two adjacent pixels. These conditions may cause the light to illuminate not only the vessel's pixel but also other nearby pixels, which also appear as high radiation value pixels on the DNB image. The CFAR detection algorithm proposed in this article can only distinguish the background pixels and the pixels lit by the vessels and cannot distinguish the specific location of the vessel target from it. We refer to the idea of Ganggang et al. [18] and use the LSD algorithm to remove adjacent pixels lit by vessel lights. The algorithm deduces the spatial proximity relation vessel of bright pixels through distance and uses the law that the brightness of the light decays with the increase in the distance to find and iteratively remove these "bright spots" of nonvessel positions to accurately extract the vessel position information from the image.

Algorithm 2 Newton Iteration Solution

Input: Maximum Likelihood Equation $[\varphi_1, \varphi_2]^T$,
Initialize: $\lambda_0 = [b_0, c_0]^T$, $k = 0$
While not converged do:
Update Φ: $\Phi_k = [\varphi_1, \varphi_2]^T_{\substack{\hat{b}=b_k \\ \hat{c}=c_k}}$;
Update A: $A_k = \begin{bmatrix} \dfrac{\partial \varphi_1}{\partial \hat{b}} & \dfrac{\partial \varphi_1}{\partial \hat{c}} \\ \dfrac{\partial \varphi_2}{\partial \hat{b}} & \dfrac{\partial \varphi_2}{\partial \hat{c}} \end{bmatrix}_{\substack{\hat{b}=b_k \\ \hat{c}=c_k}}$;
Update λ: $\lambda_{k+1} = \lambda_k - A_k^{-1}\Phi_k$;
Check the convergence conditions

$$|b_{k+1} - b_k| < \varepsilon_1, \quad |c_{k+1} - c_k| < \varepsilon_2;$$

Update k: $k = k + 1$;
End while
Output: Maximum likelihood estimation $\lambda = [\hat{b}, \hat{c}]^T$,

The input of the LSD algorithm is the detection result output by the CFAR algorithm described above, in the form of a set of bright pixels A. The implementation steps are given as follows. Select a detected highlight pixel I, and calculate the spatial distance between i and other highlighted pixels $j_1, j_2, \ldots, j_{n-1}$ respectively. Then, determine whether i and other pixels are adjacent pixels through space distance; search and find all adjacent pixels of i. The judgment condition is less than or equal to the spatial resolution of DNB data (742 m). For a pair of adjacent pixels i and j, we redefine the pixel with a larger brightness value as pixel i and delete the pixel with a smaller brightness value from A. The process repeats until the last pixel i is deleted and output when no adjacent pixel of i is found. After completing the process of output pixel i, select another pixel from the remaining pixel A, and repeat step this process until all pixels in set A are deleted; the corresponding output result is the final vessel detection result. The specific process is shown in Algorithm 3.

IV. Result

This study used the VIIRS/DNB SDR data images from two study areas to validate the proposed algorithm: the Gulf

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Algorithm 3 Local Spike Detection

Input: Bright pixel set A
Initialize: Detection result set $E = \{\ \}$
While A **is not empty:**
  Select a pixel i from set A, other pixels in A are $j_1, j_2, \ldots, j_{n-1}$
  Calculate spatial distance $d = \|i - j_k\|_2, k = 1, 2, \ldots, n-1$
  If spatial distance $d \leq 742$m
    If $(i)_{\text{brightness}} < (j_k)_{\text{brightness}}$, redefine $i = j_k$,
    End If
    Delete j_k from A
  End If
  Remove i from A and add it to E
End while
Output: Detection result E

of Mexico and the East China Sea. We selected data for the U.S. Gulf of Mexico from October 2 to 5, 2021, and the East China Sea from September 30, 2016. In order to verify the vessel detection algorithm, according to the actual situation, we combined artificial visual interpretation and AIS data for target labeling and, through some processing, to form a nighttime vessel target dataset, which will be introduced in detail.

In this article, artificial visual interpretation and AIS data are used to evaluate the detection effect of this algorithm. The visual inspection preliminarily verifies whether the automatic detection of "bright spots" in DNB images is right. However, due to the limitation of the resolution of DNB images, visual interpretation could not determine whether the "bright spots" in images are vessels or not. Therefore, the AIS data are used to further verify whether the detected "bright spots" are vessels. The AIS data of the sea area in the Gulf of Mexico are downloaded from the NOAA Office for Coastal Management, and the AIS data of the East China Sea are acquired from Elane Company.

In the actual experiment, we found that the position recorded in the AIS data does not precisely correspond to the bright spot position in the DNB image, which is mainly caused by two reasons: 1) the exact moment of the vessel target launches AIS information is not completely consistent with the moment observed by the remote sensing satellite, and there may be a time difference; 2) there are certain errors in the satellite geographic positioning system and the positioning device carried by vessels; and 3) many vessels recorded in the AIS data did not turn their lights on, resulting in not being detected as bright spots in the DNB data. Therefore, we have to combine AIS data with results from manual visual interpretation. First, visual interpretation is conducted as follows. For scattered "bright spots," we manually observe the DNB raw image and mark the "bright spots" on the image, marking a pixel as a target if it is brighter than most of the adjacent pixels in the image. Then, we simultaneously corresponded the manual marking results with AIS data records of adjacent locations, manually confirmed and marked the actual vessel position, and, finally, formed a sample set.

We adopt the method in [13] to distinguish lighting objects, such as vessels, oil platforms, and minor islands through VIIRS band M10 and the land–sea mask data. We use the VIIRS band M10 data to conduct combustion source detection and locate the position of the offshore oil platforms. We also used the land–sea mask data to find the positions of minor islands in relevant areas. Considering the limitations of random access memory (RAM) and running speed, we made a rough cut of the original image and avoided high-brightness nonvessel targets such as offshore oil platforms and minor island areas. The final dataset has 20 images of different sizes, with 879 vessels recorded by artificial visual interpretation and AIS data cross-matching. These vessels have various locations, environmental conditions, and directions, which can effectively test the algorithm's applicability.

We use precision, recall, and F_1-score to evaluate the performance of the vessel detection. Precision and recall can be expressed as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

where TP represents true positive, that is, the detected vessel is the actual target; FP stands for false positive, indicating that the detected target is not the real vessel; and FN stands for false negative and represents a real vessel that is not detected.

To be able to consider both the precision and recall of these two metrics, the harmonic average of precision and recall rate called F_1-score is also introduced as follows:

$$F_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{17}$$

The results of the two research areas and the overall detection are shown in Table II, in which the Precision value of all areas exceeds 90% and the Recall value exceeds 85%.

Fig. 4 shows a schematic of the detection results of a set of detection methods designed by us in two study areas. It can be seen that the detection algorithm can accurately detect the vast majority of vessels when the distribution of vessels is relatively dense. Among them, in Fig. 4(d) and (f), some vessels light up neighboring pixels; our algorithm can cope with this situation and output accurate detection results. The experiment proves that this method can handle complex data situations and accurately identify most vessel targets.

We implemented and tested the methods proposed by Elvidge et al. [13] and Lebona et al. [19] on our dataset and compared them with the method proposed in this article in terms of detection performance. Elvidge's method is designed for DNB data and has wide influence. Based on this method, a large number of vessel target detection products have been produced and applied to fishing vessel detection and other fields [10]. The method proposed by Lebona et al. [19] is a two-parameter CFAR, while the core of the method proposed in this article is based on the idea of a CFAR detector, which is mainly improved in sample selection, distribution parameter estimation, and postprocessing procedure. It is obvious that these two methods have similar ideas, and the actual effect of the improvement measures proposed in this article can be

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TABLE II
CALCULATION RESULTS OF PERFORMANCE ASSESSMENT INDEX

Method	Study area	Precision	Recall	F1 score
Our method	Gulf of Mexico	0.9527	0.9430	0.9478
	East China Sea	0.9167	0.9026	0.9096
	total	0.9448	0.9340	0.9394
SMI Threshold method	Gulf of Mexico	0.8799	0.8567	0.8681
	East China Sea	0.8360	0.8103	0.8229
	total	0.8702	0.8464	0.8581
Two-parameter CFAR	Gulf of Mexico	0.9346	0.8977	0.9157
	East China Sea	0.8913	0.8410	0.8654
	total	0.9251	0.8851	0.9047

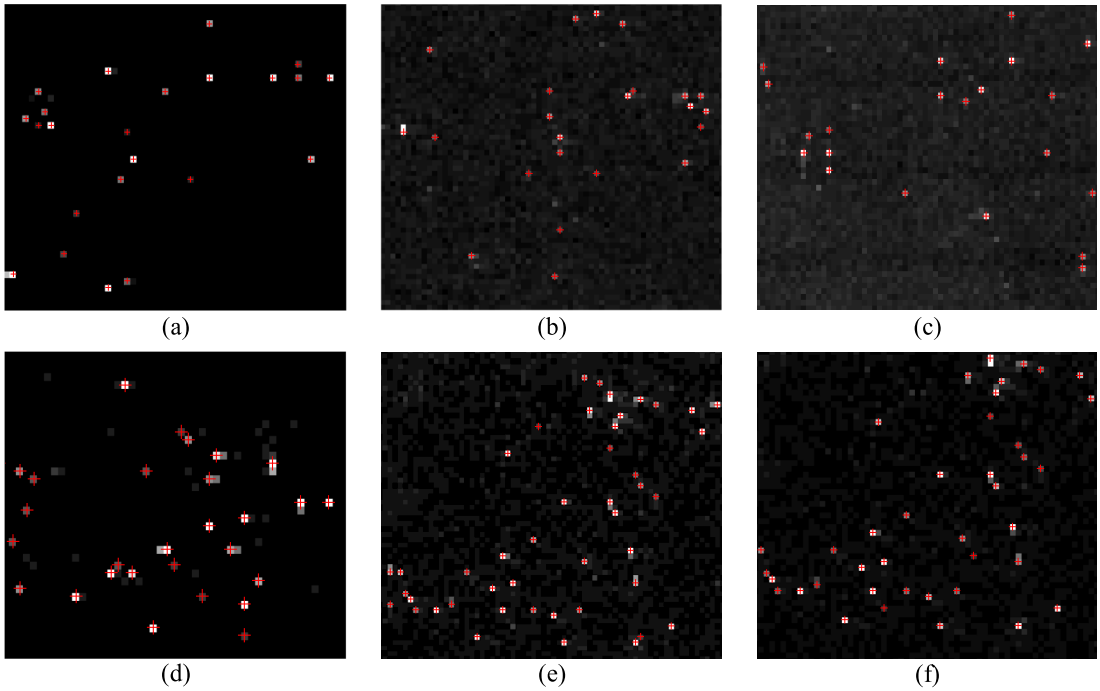


Fig. 4. Schematic of the detection results in two study areas. (a)–(c) Data from the Gulf of Mexico. (d)–(f) Data from the East China Sea.

reflected directly through comparison. Therefore, we choose these two methods for comparison.

The detection results of the comparison method are shown in Table II and Fig. 5. It is obvious that the detection precision and recall value of the SMI threshold method are lower than those proposed in this study, which can be attributed to the fact that the SMI threshold method is a fixed threshold method. Although researchers have designed complex and relatively compelling preprocessing and postprocessing steps, the fixed threshold designed based on manual experience still cannot adapt to the complex and changeable environmental conditions in different regions, and the wrong detection may occur in one part of the region, while the missing detection may occur in another part. The two-parameter CFAR method is based on the statistics and modeling of the local background to determine the threshold value, which is better than the fixed threshold method with higher detection accuracy and recall rate. However, this method has a small number of wrong detection and a large number of missed detection cases in areas with dense vessel lights. The main reason is that the pixels illuminated by the vessel's light in these regional candidate boxes affect the step of parameter estimation and the selection of threshold values, resulting in poor detection performance.

V. DISCUSSION

It can be seen from the experimental results that compared with the other two methods, our method achieves a better detection effect in complex detection tasks and successfully detects the vast majority of vessel targets. A small number of false positives and false negatives occur, mainly related to the following factors. First, the selection of cutoff points in candidate box samples may sometimes be inaccurate, resulting in inaccurate threshold selection. In addition, due to the low spatial resolution of DNB data, there is a situation in which a highlighted pixel contains two vessels in an area with dense vessel distribution, and the adjacent highlighted pixel may lit by one vessel or two vessels, which is difficult for any detection methods to judge at present.

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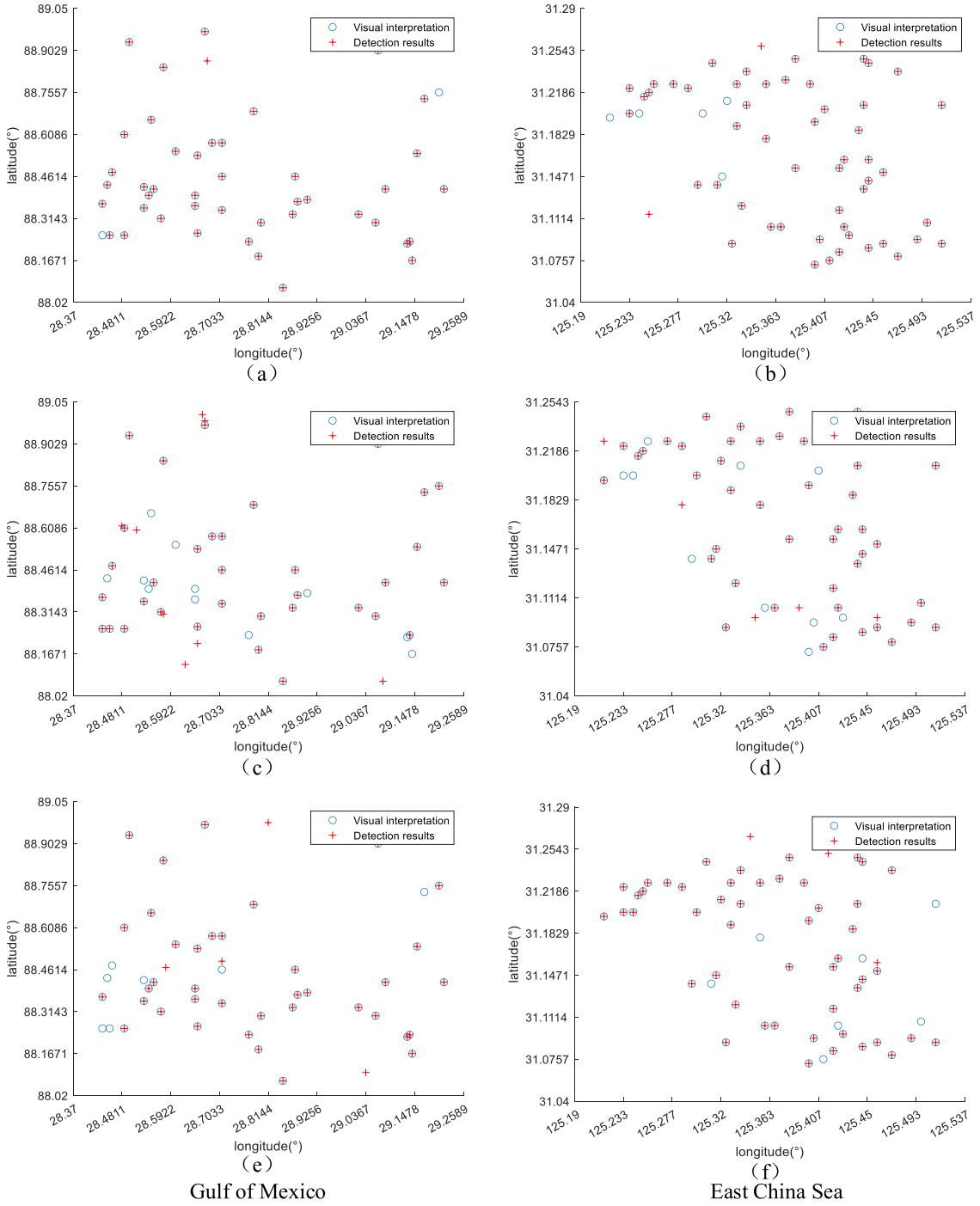


Fig. 5. Validation results were obtained through selections of visual interpretation and AIS data cross-matching. (a) and (b) Detection result of our method, (c) and (d) SMI threshold method, and (e) and (f) two-parameter CFAR.

It is obvious that there are still potential research values and prospects in nighttime vessel detection. The basic principle of vessel detection based on low-light nighttime remote sensing data is to make use of the lights carried by the vessel. If the lights carried by the vessel are not turned on, the vessel position reported by AIS data cannot correspond to the light position in DNB data, and it will not be recorded in the dataset; otherwise, the algorithm cannot be verified correctly. Therefore, we expected to design algorithms based on multisource data fusion in future work to solve the detection problem of unlit vessels at night. Also, for DNB data with lower spatial resolution, in areas with dense distribution of vessels, there are two vessels in one highlight pixel, and whether the adjacent highlight pixel is lit by one vessel or two is difficult to determine. Therefore, in the future, it is expected that the improved algorithm can realize the estimation of the number of vessels in the local area. We also hope to adapt our method to high-resolution low-light remote sensing images to obtain more accurate detection results.

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VI. Conclusion

In this article, a method based on a CFAR detector is proposed for the automatic detection of vessels at night with VIIRS DNB data. The algorithm considers the influence of interference items in areas with dense vessel distribution on threshold selection and complex situations, such as the vessel target lighting up multiple pixels, and can output accurate vessel detection results. First, the MVI is designed and utilized to iteratively achieve the truncation depth of the samples and ultimately obtain the undisturbed background samples and the adjusted Weibull distribution. Subsequently, the Weibull distribution was adjusted according to the sample truncation depth, and the background sample was modeled. The high-light pixels were obtained by calculating the threshold of the false alarm rate. Finally, the LSD algorithm is used to remove the nonvessel bright pixels and, finally, obtain the vessel detection results.

In order to test the algorithm's performance, we chose two research areas, the East China Sea and the Gulf of Mexico, and built a vessel target dataset. We annotate vessel targets through AIS data combined with artificial visual interpretation as a reference to verify the performance of the algorithm. We applied the algorithm to this dataset, and the results show that the total detection accuracy and recall rate of the algorithm in all research fields are 94.48% and 93.40%, respectively. The algorithm overcomes complex environmental factors and image noise and achieves good results, which shows that the algorithm is reliable.

In future work, we plan to further explore the relevant characteristics of the vessel light source, hoping to improve the performance of the algorithm. In addition, we expect more validation work to be carried out to analyze the types of the detected vessels.

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